

SYSTEM DESIGN DOCUMENT FOR

***Implementation of RGB Dimensionality
Reduction Techniques using Denoising
Autoencoder***

Prepared by

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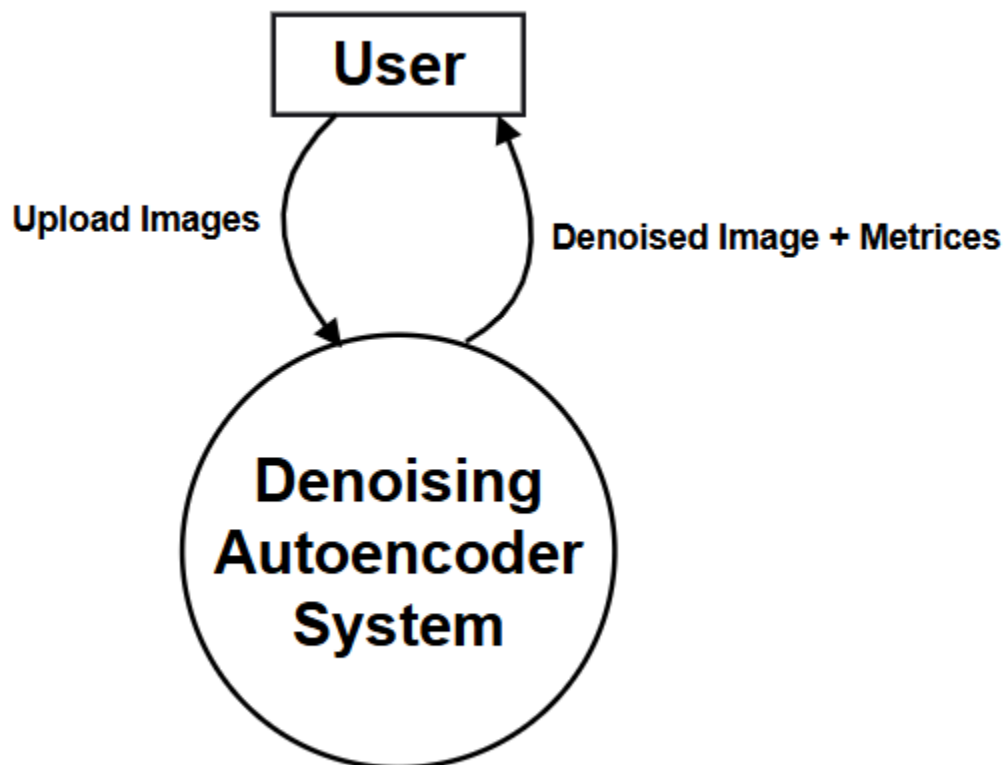
***DEPARTMENT OF INFORMATION TECHNOLOGY
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SECTION: DATA FLOW DIAGRAMS

Level 0 : Context Diagram

Description:

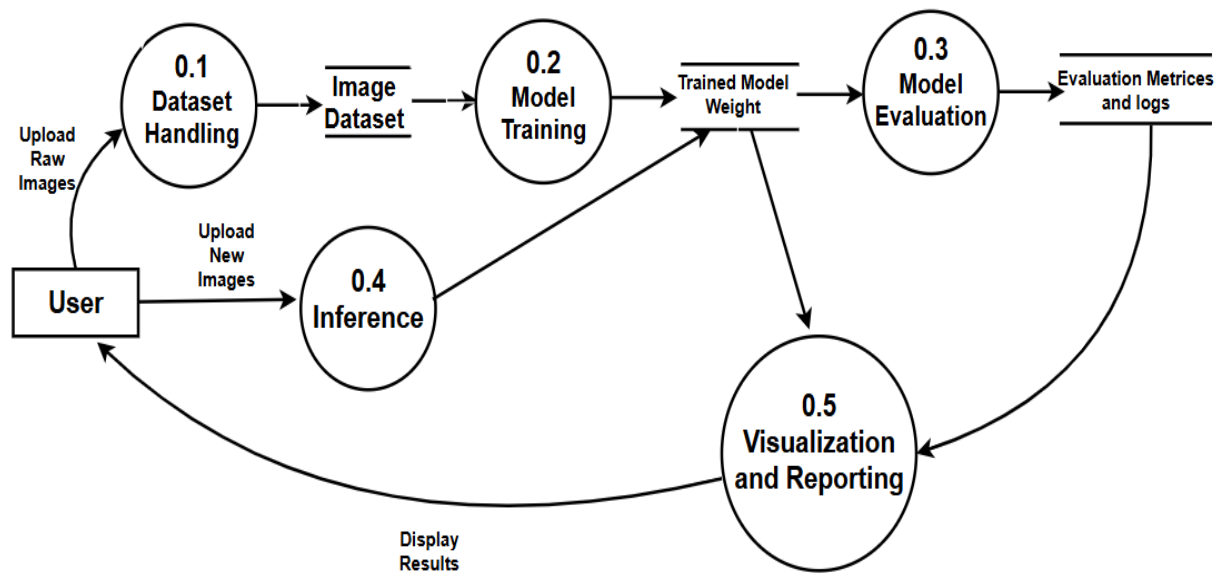
The level 0 DFD represents the Denoising Autoencoder System as a single process, showing its interaction with the external entity.



Level 1 : Top-Level Decomposition

Description:

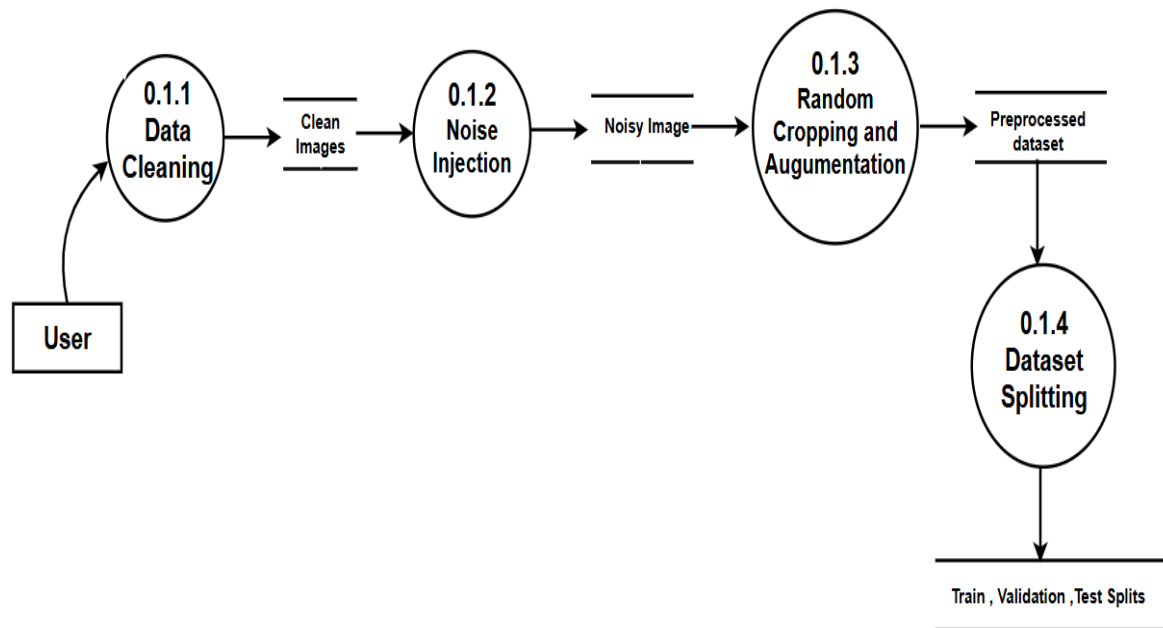
The Level 1 DFD decomposes the system into its major sub-processes: preprocessing, training, evaluation, storage, and inference.



Level 2 : Dataset Handling

Description:

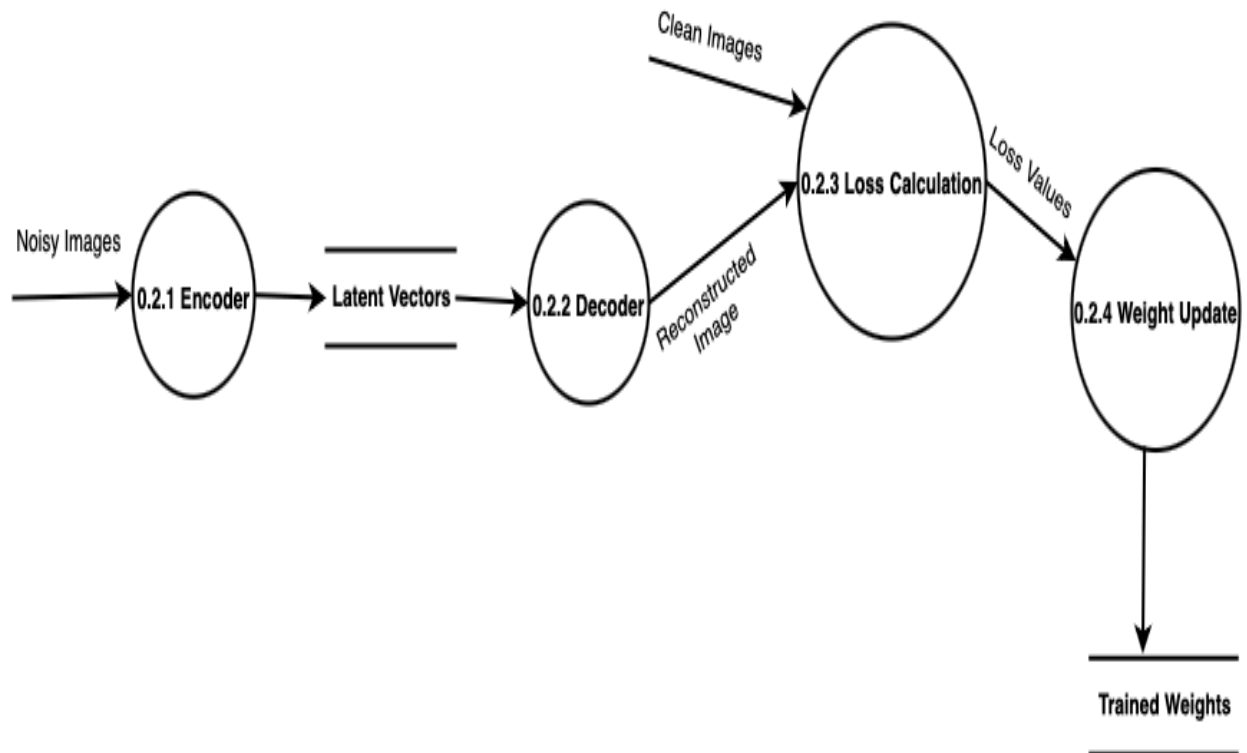
The Level 2 DFD expands the Dataset Upload and Preprocessing module. It manages dataset storage, noise injection, and augmentation.



Level 3: Model Training

Description:

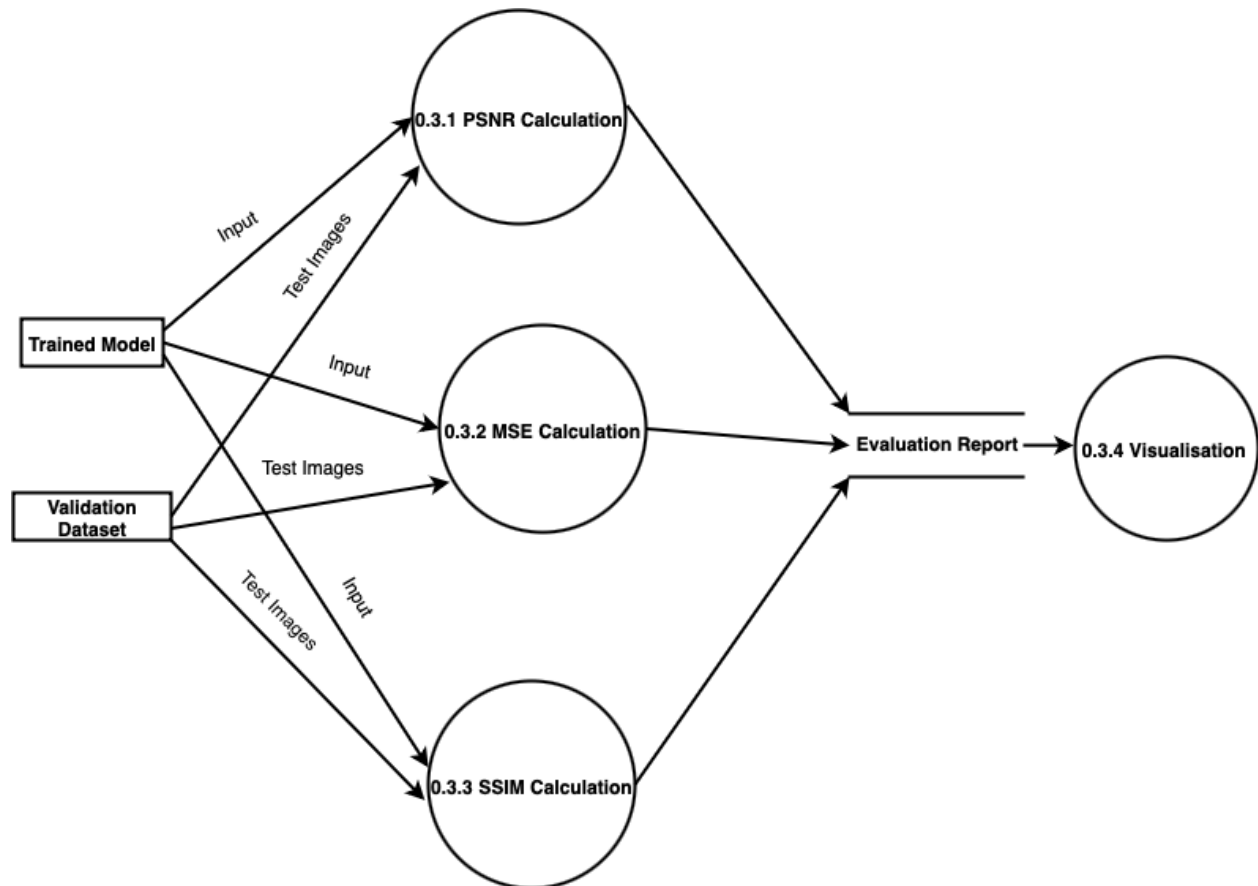
Level 3 elaborates the Model Training process with encoder-decoder structure and iterative optimization.



Level 4: Evaluation and Visualization

Description:

Level 4 details model evaluation and visualization of results.



Data Dictionary

Data Stores

1.Dataset Store:Repository of clean, noisy, and augmented datasets used for training.

2.Trained Model store:Stores the trained autoencoder model weights and configurations.

Data Flows

1.Uploaded Dataset

- Source: User → Data Handling
- Description: Raw RGB image dataset uploaded by the user.

2.Noisy Dataset

- Source: Noise Injection → Preprocessing
- Description: Dataset after Gaussian noise is applied.

3.Augmented Dataset

- Source: Cropping/Augmentation → Training
- Description: Dataset after preprocessing and augmentation.

4.Denoised Images

- Source: Inference → User
- Description: Reconstructed clean images after passing through the autoencoder.

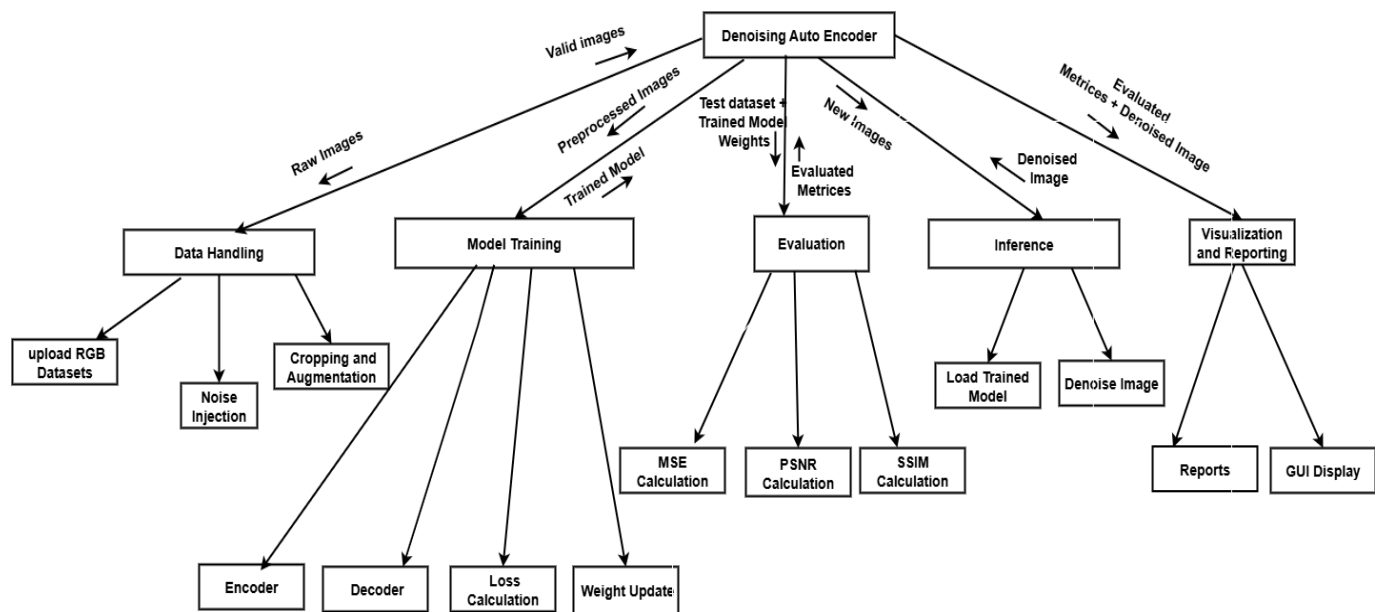
5.Evaluation Metrics

- Source: Evaluation Module → GUI/Reports
- Description: Metrics used to evaluate reconstruction quality.

SECTION: STRUCTURAL CHART

Description:

The structural chart represents the hierarchical breakdown of the Denoising Autoencoder System. It shows the major functional modules and their submodules, along with the flow of control between them.



Contributions

The development of the **Denoising Autoencoder System** was divided among team members based on major functional modules. The contributions are as follows:

1.Data Handling and GUI Development - Ankita A(231IT007)

- Implement dataset upload, noise injection, and augmentation mechanisms.
- Design and develop the graphical user interface (GUI) for user interaction and visualization of results.

2.Model Training - Peteti Hanvitha (231IT048)

- Implement the encoder-decoder architecture.
- Manage storage and retrieval of trained models.

3.Part of Model Training,Evaluation and Visualization- Nelluri Thanmaya Sri (231IT0043)

- Develop the training pipeline including loss calculation and iterative weight updates.
- Design evaluation modules for computing MSE, PSNR, and SSIM.
- Develop visualization components including performance curves, metrics display, and reporting.